

P100-056 - Calibrated timber-concrete beam response

Scenario and task

Three native workbooks contain six composite-beam summary curves, interface-slip summaries and detailed strain records for one beam. This task uses all eligible late summary-curve points from normal-weight and lightweight concrete beams A/B/C. Post-damage force drops are retained; native source bytes are unchanged.

Experimental and validation design

The target is applied force F in kN conditional on imposed deflection d in mm. Each beam provides a calibration prefix ending before its first deflection above 10 mm. F_0 and d_0 are its final calibration point, $k_0 = F_0/d_0$. Later force, failure load and future calibration are forbidden. Four A/B beams support complete-beam development folds; two C beams provide 45 reserved late points. Two beams are insufficient for population confidence claims.

At least five adaptive attempts preceded a code-and-state freeze; confirmation was then evaluated without fitting or reselection. Source publications and supplied analyses are known. All packaged outcomes are now exposed to later users.

Frozen equation and interpretation

Define $u = d - d_0$, $k_0 = F_0/d_0$, and $m = 1$ for lightweight concrete ($m = 0$ for normal-weight concrete). The frozen selected reference is

$$\hat{F} = \min \{F_0 + k_0 u, C_0 + C_1 m\}, \quad C_0 = 60.247645 \text{ kN}, \quad C_1 = -5.080368 \text{ kN}.$$

The elastic component uses only the initial calibration prefix. The cap approximates loss of composite-action stiffness; it is not a design resistance. Exact coefficients and all competing states are in rules.json.

Evidence and limitations

The primary metric is the equally weighted mean of within-group mean absolute errors in kN; sample counts do not create independent replicates.

Model	Development MAE	Confirmation MAE
Elastic continuation control	22.3328	14.7367
Quadratic response control	5.64735	8.73499
Progressive compliance	4.81833	9.67599
Common force cap	3.72692	4.75397
Concrete-specific cap (selected)	3.6113	5.28109
Post-yield response	3.72897	5.13079
Recent-tangent calibration	4.9214	9.28458

Selected-reference group errors: P-LWC-C: 5.52815; P-NWC-C: 5.03402.

The main supported conclusion is protection against unrestricted elastic continuation. Concrete-specific advantage is not confirmed: the simpler common cap achieves 4.75397 kN, better than the preselected 5.28109 kN. The reference is not replaced after seeing this. Progressive rational compliance and the recent-stiffness variant achieve 9.67599 and 9.28458 kN, respectively; their favorable development fit fails to transfer to both C beams. Abrupt force drops and specimen variability remain important counterexamples.

Practical meaning and reproducibility

Saturation protects against gross elastic extrapolation and gives a compact calibrated-response benchmark. It is not a structural design limit: the fitted cap is neither measured characteristic strength nor a safety-rated allowable load. The task needs initial beam loading and does not predict ultimate strength from virgin material specifications.

Run python run.py to check hashes and reproduce every saved equation, prediction and metric. rules.json holds all coefficients, parameter names and fit scope; findings.json separates supported claims, failed hypotheses and abstentions. The shared benchmark evaluator accepts alternative equations under this same information budget; exact agreement with this reference is not required.

Sources

[Author manuscript](#)

[Source publication](#)

[Zenodo measurement workbooks](#)